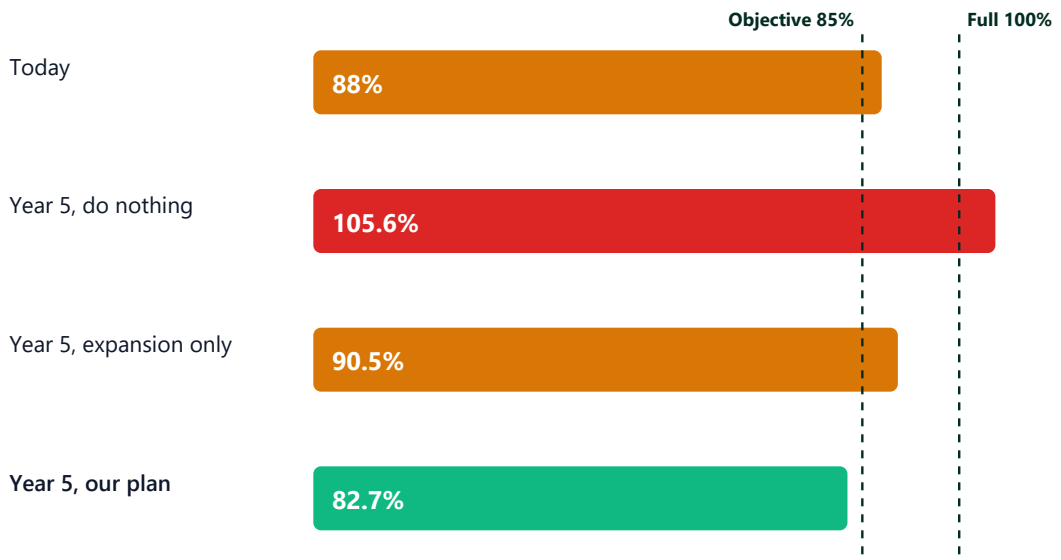
THE CHALLENGE

PortFlow's problem, and why it underperforms

PortFlow's current operating model is struggling to accommodate growing container volumes, resulting in excessive yard congestion, inefficient container movement and delayed truck processing.

88%	6 days	72 min	$14M/yr
YARD UTILIZATION TODAY	AVERAGE DWELL	TRUCK TURN TIME	OVERTIME + PENALTIES
21,120 of 24,000 TEU	every box held 6 days	40 queue + 32 inside	$6M + $8M

YEAR-5 YARD UTILIZATION AT +20% VOLUME



Year-5 objectives: yard utilization <= 85% and truck turn < 45 minutes. Today PortFlow meets neither.

WHY IT UNDERPERFORMS: ONE CAUSE, THREE MECHANISMS

Fragmented release and schedule information

Release status, vessel changes and pickup plans are not shared across teams, truckers, rail or customers; planning runs in separate systems on fixed rules.

Containers wait
Customers learn late that a box is released, so pickups slip
Dwell 6 days

Trucks bunch
No appointments or readiness signal, so arrivals cluster
40 min queue

Stacks get reshuffled
Exact position and pickup time unknown, so cranes dig
32 min inside

SYMPTOMS MANAGEMENT SEES

Yard crowding	72-min truck turns	$6M overtime
$8M penalties	Customers can't plan	Staff anxiety

THE PROOF, FROM THE CASE DATA

Little's Law: inventory = throughput x dwell. 3,520 TEU/day x 6 days = 21,120 TEU. At 4,224 TEU/day the same dwell needs 25,344 TEU (106%). The 15-acre parcel only reaches 90.5%. Dwell must fall to 4.83 days or less.

RECOMMENDED PACKAGE

Integrated flow package: shared visibility, appointments, AI-assisted yard planning

PILLAR 1
See it: shared release and slot-level visibility

Release and ETA visibility portal — **$1.4M**

One shared status for lines, truckers, rail and customers, with customs-hold flags and a named escalation contact.

EXPECTED EFFECT
Release status, holds and exact container position visible to every party

PILLAR 2
Schedule it: truck appointments and dwell rules

Truck appointment system with gate pre-advice and OCR — **$2.2M**

Spread arrivals into booked slots, only for released and ready containers; faster gate transactions.

Dwell-time rules: free-time tariff and empties evacuation — **$200K**

Storage charges after free days, early-pickup incentives and agreements with lines to remove empties.

EXPECTED EFFECT
Queue 40 to about 14 min; dwell 6.0 to 4.7 days

PILLAR 3
Plan it together: AI-assisted yard plan and one operations cell

Slot-level positioning and AI-assisted stacking — **$2.9M**

RTG position sensors plus a planning upgrade so boxes due out first sit on top; planners approve each recommendation.

Integrated operations planning cell — **$800K**

One shared plan across vessel, yard, gate and rail teams, so a schedule change is re-planned once, not in several systems.

EXPECTED EFFECT
Fewer rehandles; time inside 32 to about 26 min

ENABLERS

Change management and paid training — **$700K**
Co-design workshops, super-users per shift and training in paid working time.

Contingency (about 10%) — **$800K**
Released at the phase-3 gate if unused.

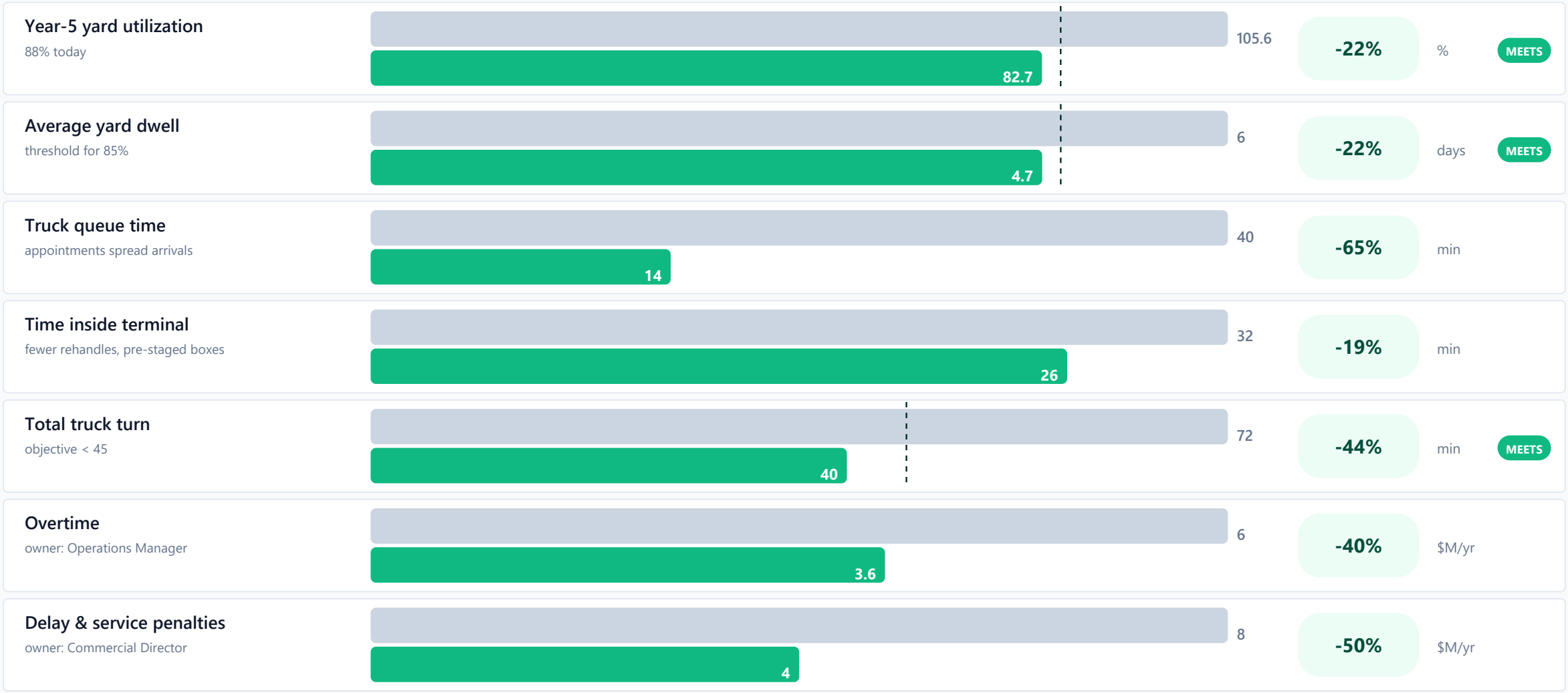
$9.0M of $10.0M upfront · $1.3M/yr to run · every saving has a named owner

Set aside: Expansion only: about $14.5M, over budget, and still 90.5% utilization · Full crane automation: far above $10M, labour and safety risk, and no fix for the information gap

KPI IMPACT AT +20% VOLUME

From congestion to flow: every year-5 objective met

Grey = today · green = with the package · dashed line = year-5 objective. All values at +20% volume.



BUDGET ALLOCATION

Budget allocation and decision matrix

WHERE THE $10.0M GOES



Truck appointment system with gate pre-advice and OCR	$2.2M	22%
Release and ETA visibility portal	$1.4M	14%
Slot-level positioning and AI-assisted stacking	$2.9M	29%
Integrated operations planning cell	$800K	8%
Dwell-time rules: free-time tariff and empties evacuation	$200K	2%
Change management and paid training	$700K	7%
Contingency (about 10%)	$800K	8%
Total committed	**$9.0M**	
Unspent headroom	$1.0M	

83% of the spend goes straight at the core issue.
See it: $1.4M
Schedule it: $2.4M
Plan it together: $3.7M
Enablers: $1.5M
Running cost $1.3M a year: software, support, sensors, analysts and refresher training.

DECISION MATRIX (SCORE /10; BUDGET IS A HARD GATE)

	A · Gate quick win: truck appointments and release portal	B · Integrated flow package: shared visibility, appointments, AI-assisted yard planning	C · Yard expansion on the 15-acre parcel with two added RTGs
Target attainment (30%)	3.4	7.4	4.9
Financial return (NPV) (25%)	9.1	10.0	2.0
Payback speed (15%)	5.2	4.7	0.0
Downside resilience (-15% perf.) (15%)	4.4	9.5	6.2
Budget headroom (15%)	5.9	1.0	0.0
Weighted score	5.63	7.0	2.9
Within budget	yes	yes	NO
Meets year-5 objective	no	yes	no
5-year net impact	-$9.4M	-$16.5M	$13.5M

Option B is the only direction that meets the objective inside the budget, and it has the best five-year impact.

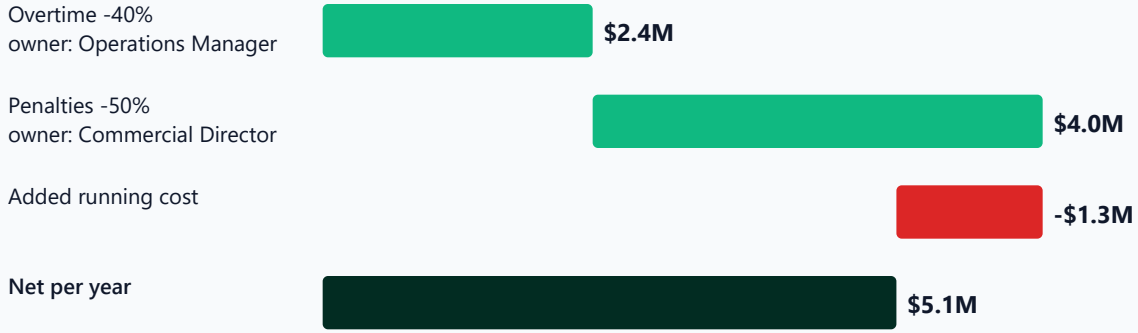
FINANCE

The financial case

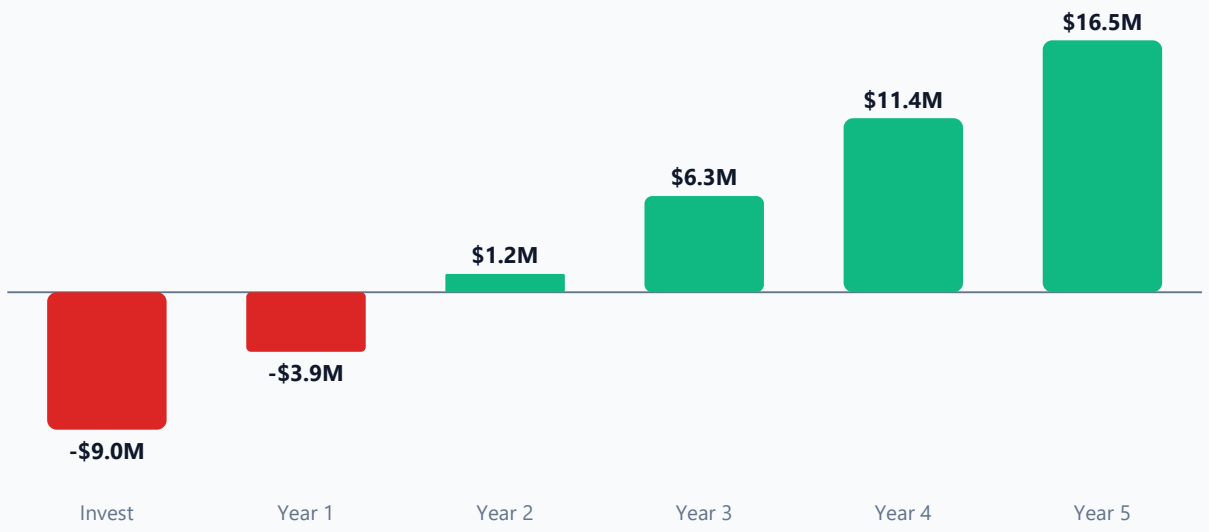
FIVE-YEAR NET FINANCIAL IMPACT = INVESTMENT + 5 X (ADDED ANNUAL COSTS - ANNUAL SAVINGS)

$9.0M + 5 x ($1.3M - $6.4M) = -$16.5M (net savings)

ONE OPERATING YEAR



CUMULATIVE POSITION, UNDISCOUNTED (CASE METHOD)



FINANCIAL UNCERTAINTY

How much of the penalty bill comes from customs or documentation holds that PortFlow cannot override. If penalties fall only half as much, savings are $4.4M and the impact is still -$6.5M. Break-even needs only $3.1M a year of savings; the investment is recovered after 1.8 operating years. Discounted cross-check: NPV $6.5M at 9.5%.

12 MONTHS, THREE GATED PHASES

Implementing without disrupting operations



HOW OPERATIONS KEEP RUNNING

- Lane by lane and block by block: never more than 2 gate lanes or 1 yard block changing at once
- Go-lives scheduled outside vessel peak windows; sensors fitted on off-peak shifts
- Parallel running: old rules stay available until the new process passes its gate
- Rollback plan: if gate throughput drops below baseline for 2 shifts, revert that lane

MARKET CONTEXT

Market analysis: what the industry tells us

Demand is rising across Canadian gateways, peers win on reliability rather than space, and automation is a labour flashpoint.

MARKET FORCES

Volumes are growing [1]

The Port of Vancouver handled a record 3.8M TEU in 2025, about 9% above 2024, as Canada diversified trade. PortFlow's +20% over five years is in line with this.

Congestion is the industry's pain point [2]

Container dwell at Canada's largest port was still a significant issue in early 2025, with rail capacity lagging the surge in boxes.

Customers buy reliability [0]

Shippers and truckers plan around release certainty. Terminals that share status and keep appointment windows win repeat volume; uncertainty shows up as penalties.

Automation is a labour flashpoint [5]

The November 2024 B.C. port lockout centred on automation and manpower; Ottawa ordered binding arbitration on both coasts.

WHAT PEER TERMINALS DO

GCT Bayonne (NY/NJ)	Truck appointment system, phased rollout with trucker outreach	**Turn times in appointment hours improved by more than 40%; over 90% of reservations kept**	[3]
LA / Long Beach terminals	Appointment systems introduced in 2015	**Mixed results: no clear turn-time gain without wider coordination**	[4]
Seattle, Houston, Tacoma	Extended-dwell fees after free time	**Escalating daily charges (e.g. Houston $50 to $150 per unit per day) push import boxes out of the yard**	[6]

SO WHAT FOR PORTFLOW

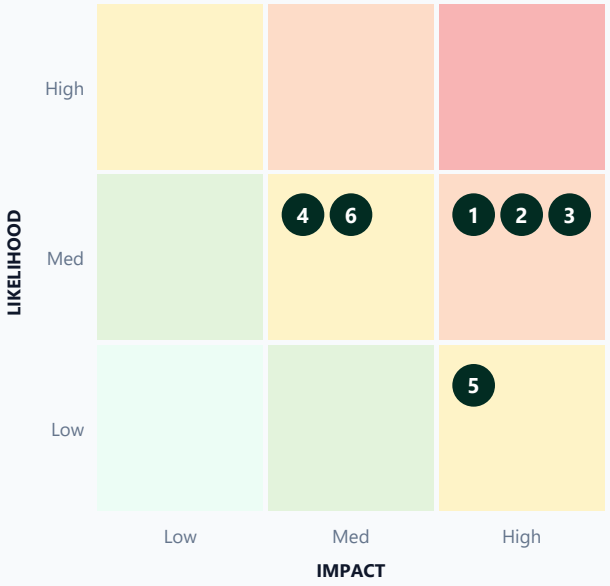
- Appointments work only with readiness information and partner buy-in, which is why PortFlow pairs them with the release portal and co-design.
- Dwell pricing is standard practice, so a free-time tariff is defensible to customers.
- Choose AI that assists people over crane automation, and pair it with a no-layoff commitment, to avoid the dispute seen at B.C. ports.
- Reliability, not more space, is how PortFlow defends its share as rival gateways add capacity.

Sources: [1] Port Technology, Port of Vancouver posts record 170m metric tonnes in 2025 [2] Trains.com, CPKC container dwell still an issue at Canada's largest port [3] US EPA Ports Initiative, GCT Bayonne drayage truck appointment system [4] AJOT, Do truck appointment systems decrease turn times? Preliminary data shows no [5] CBC, B.C. ports are frozen amid a labour dispute [6] gCaptain, Port Houston to begin charging for long-dwelling import containers; Green Worldwide (Seattle); FreightWaves (Tacoma) [0] Case brief: stakeholder perspectives (trucking, supply chain partner)

EMPLOYEES, AI, PARTNERS, MONEY

Key risks and mitigation

RISK MAP: LIKELIHOOD X IMPACT



Risk budget: $800K contingency held by the Finance Director, plus $700K upfront and $150K/yr for change management and paid training. Each risk has an owner and an early-warning trigger reviewed at the monthly steering meeting.

#	RISK	MITIGATION	OWNER & FUNDING	EARLY WARNING
1	**Truckers do not adopt appointments** **Medium / High**	Slot rules co-designed with trucking firms, grace windows, priority lanes for on-time trucks, a walk-in lane kept during transition	Gate and Landside Manager Appointment system $2.2M (co-design included)	Slot adherence < 75% for 2 weeks
2	**Dwell does not fall enough** **Medium / High**	Free-time tariff, release alerts, empties evacuation agreements; parcel design kept ready as phase 2	Commercial Director Tariff design $0.2M; parcel design held in contingency	Dwell > 5.3 days at month 9
3	**Cutover disrupts live operations** **Medium / High**	Lane-by-lane and block-by-block rollout, parallel running, go-lives outside vessel peaks, rollback plan	Operations Manager Contingency $0.8M	Gate throughput below baseline for 2 shifts
4	**Staff resistance and role concerns** **Medium / Medium**	Staff involved before tools arrive, paid training, no layoffs from this program, redeployment into the planning cell	HR Director with the Employee Representative Change and training $0.7M + $0.15M/yr	Training completion < 90% or AI advice acceptance < 50%
5	**AI advice wrong or data quality poor** **Low / High**	Advisory only, 6 weeks in shadow mode, hard safety rules locked, weekly model review	Yard Planning Manager Positioning and AI $2.9M, shadow mode included	Override rate > 30% or any near-miss linked to a recommendation
6	**Savings fall short (customs-driven penalties)** **Medium / Medium**	Holds flagged early in the portal; savings tracked monthly by owner; plan still pays back at $3.1M/yr	Finance Director No extra spend; break-even $3.1M/yr	Penalties down < 25% by month 12